Deccan International School

ICSE 2024-25
High School
First Term Examination

Subject: Physics

Date: 20-09-2024 **Maximum Marks: 80**

Grade:8 ICSE **Duration: Two hours**

Name: **Roll no:**

Instructions:

1) Answers to this Paper must be written on the paper provided separately.
2) You will not be allowed to write during first **15** minutes
3) This time is to be spent in reading the question paper.
4) The time give at the head of this paper is the time allowed for writing the answers.
5) ***Section-A*** is compulsory. Attempt ***all*** questions from ***Section-B***.
6) The intended marks for questions or parts of questions are given in brackets [ ].

SECTION – A

*(Attempt **all** questions from this section.)*

Question 1

Choose the correct answers to the questions from the given options. **[15]**

(Do not copy the question, write the correct answer only.)

(i) If the distance between the force and the point of rotation doubles, the moment of force:
(a) Halves (b) Doubles
(c) Remains the same (d) Becomes zero

(ii) Atmospheric pressure decreases with:
(a) Decreasing altitude (b) Increasing altitude
(c) Increasing temperature (d) Decreasing temperature

(iii) A worker pushes a box across the floor with a constant force of 50 N, but the box doesn't move. Which force is responsible for preventing the box from moving?
(a) Gravitational force (b) Frictional force
(c) Centripetal force (d) Magnetic force

(iv) A diver experiences more pressure as they go deeper underwater because:
(a) The density of water increases with depth (b) The temperature decreases
(c) The force of gravity increases (d) The weight of the water above them increases

(v) In the given figure below both teams pull the rope with equal force, yet it doesn't move. What is this condition called?



(a) Balanced forces
(b) Unbalanced forces
(c) Gravitational equilibrium
(d) Elastic force

(vi) A balloon filled with air is placed in a freezer. After a while, the balloon shrinks. This is because:
(a) The particles of air freeze
(b) The pressure inside the balloon increases
(c) The gas particles lose energy and move closer together
(d) The balloon loses air

(vii) Which of the following will increase the pressure exerted by a force on a surface?
(a) Increasing the surface area
(b) Decreasing the surface area
(c) Reducing the applied force
(d) Increasing the volume of the object

(viii) In the given figure below, a cyclist needs more energy to ride uphill because;



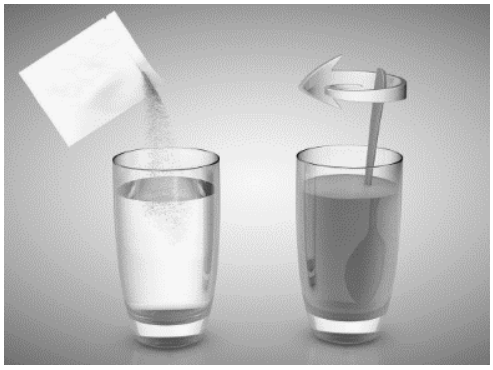
(a) The mass of the cycle increases
(b) The gravitational potential energy increases as the height increases
(c) Friction decreases
(d) Wind resistance increases

(ix) A piece of wood floats on water. The buoyant force on wood will be:
(a) Zero
(b) more than the weight of the wood piece.
(c) equal to the weight of the wood piece
(d) less or more than the weight of the wood piece.

(x) The S.I. unit of work is:

(a) joule (b) newton (c) calorie (d) watt

(xi) Observe the given figure below. When the sugar is added into the water and stirred it for sometime, the sugar gets dissolved. What happens to the mass of the water after dissolving?



(a) The mass of the solution decreases (b) The mass of the solution increases

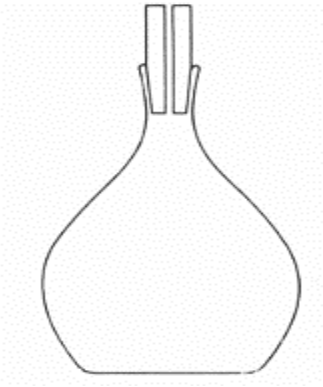
(c) The mass remains the same (d) Sugar loses mass

(xii) What is the correct order, in increasing size, for measurement of length?

(a) kilometre, centimeter, meter (b) meter, centimeter, millimeter.

(c) millimeter, centimeter, meter (d) millimeter, meter, centimeter.

(xiii) Identify the instrument given below:



(a) Cylindrical beaker (b) density bottle (c) Cylindrical vessel (d) thermal flask

(xiv) The force of attraction produced between molecules of two dissimilar type of objects are called as_____

(a) Cohesive force (b) electrostatic force (c) adhesive force (d) frictional force

(xv) At the same height, if a mass of the body is doubled, its potential energy:

(a) Increases by two times (b) increases by four times

(c) Decreases by two times (d) remains unaltered

Question 2

(i) ***State whether the following statements are true or false. Correct the false statements:*** **[5]**

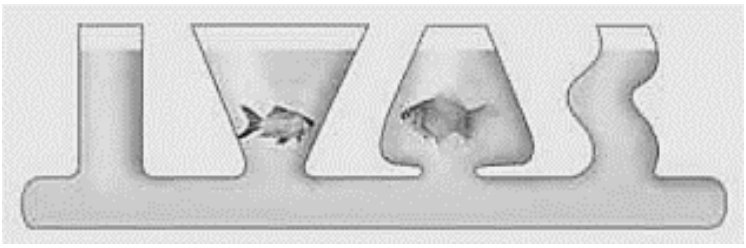
1. When matter changes its state there is no change in temperature.
2. The unit of relative density is kilogram per cubic meter.
3. Force can make the object to turn if it is not attached to hinge.
4. Liquid exerts pressure only in downward direction.
5. Potential energy depends on the speed of the moving body.

(ii) ***Explain the following terms:*** **[5]**

1. One joule
2. Atmospheric pressure
3. Deposition
4. Density
5. Pressure

Question 3

(i) Study the picture below and answer the following questions:



1. Is the pressure at a depth ***h*** is same or different for different containers? Give reason to justify your answer.
2. If the liquid is replaced with the oil to the same height, will the pressure exerted by the liquid remains same? **[3]**

(ii) The mouth of the inflated balloon has to be closed. What happens if the mouth of the inflated ballon is opened and why? **[2]**

(iii) Ice floats in water but sinks in cooking oil. Give ***two*** reason to justify your answer. **[2]**

(iv) Porters place a round piece of cloth on their head when they have to carry heavy load. Give reason. **[2]**

(b) ***Surface of snow turns into steam without melting***. Name the phenomenon and define it. **[2]**

(c) Observe the given figure and answer the following questions:



(a) What is the magnitude of work done in the given figure?
(b) Give reason to justify your answer in part(a) **[2]**

(d) State the factors affecting the moment of force. **[2]**

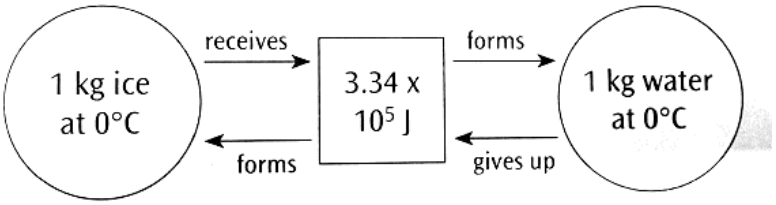
SECTION – B

*(Attempt **all** questions from this section.)*

Question 4

(i) (a) Explain the action of rubber sucker based on atmospheric pressure.
(b) What is the gravitational unit of force? **[3]**

(ii) Observe the given figure and answer the following questions:



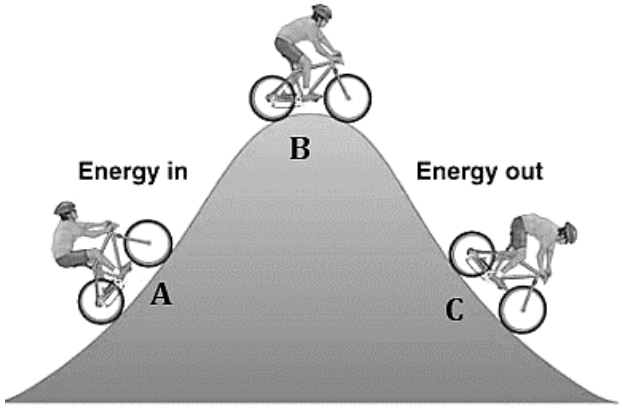
(a) When ice changes into water, which phenomenon takes place?
(b) What change in temperature is seen when the phenomenon in part(a) takes place?
(c) What is the quantity of heat absorbed by the ice to change its state called? **[3]**

(iii) The mass of iron ball is 800g. What should be the volume of the ball if the density of iron is $7 gcm^{-3}$? Also find the :

(a) density of iron in SI unit
(b) relative density of iron. **[4]**

Question 5

(i) Observe the given figure and answer the following questions:



(a) At which point the cyclist possess maximum potential energy?
(b) State the factor on which the maximum potential energy gained in part(a) depend upon.
(c) Write the energy conversion that takes place when cyclist moves from point B to point C **[3]**

(ii) (a) ***Relative density of iron is 2.7***. What is the meaning of the statement?
(b) Does the density of substance change with change in shape or size of an object?
(c) Name the energy possessed by compressed spring. **[3]**

(iii) (a) Reena applies a force of 1.5N on the handle of the door of width 1.2m to open it. Calculate the minimum moment of force required to open the door.
(b) A normal force of 100N can produce a pressure of 10000Pa. Calculate the area on which the force shall act to exert the pressure. **[4]**

Question 6

(i) Explain with a neat diagram that liquid pressure increases with the increase in height of the liquid column. **[3]**

(ii) (a) Distinguish between thrust and pressure (*any one point*).
(b) What will be the magnitude of moment of force acting on the object if the force is at the pivot point? Give reason to justify your answer. **[3]**

(iii) A force of 50 N is applied to a surface area of 0.25 m^2. Calculate the pressure exerted on the surface. If the same force is applied to a smaller area, explain how the pressure changes and why. **[4]**

Question 7

(i) An empty specific gravity bottle weighs 32g in a physical balance. When it is completely filled with water, it has a mass of 72g. On filling it with methanol, the mass of the bottle is found to be 64g. Calculate the relative density of methanol. **[3]**

(ii) (a) Why do we feel cool on standing under a fan while sweating?
(b) Give one example on which rate of evaporation increases with increase in temperature **[3]**

(iii) A person lifts a 6 kg bucket from the ground to a height of 1.5 meters, then places it on a table 0.5 meters higher. Calculate the potential energy at both heights. (Take g =10Nkg^{-1}).
What happens to potential energy when the bucket is taken from 1.5m and kept it on a table of 0.5m higher? **[4]**